



EcoTrace
TRACK

SECURE CHAIN OF CUSTODY

Smart Biomedical Waste Tracking & Compliance Platform

Automated barcode generation, integrated mobile weighing, and real-time digital auditing for modern healthcare facilities.

Product **V1.2**

Focus **Compliance & Security**

Powered by **Cloud Integrations**

Reimagining Waste Logistics

Traditional biomedical waste workflows suffer from high manual overhead, compliance risks, missing data, and critical auditing delays. EcoTrace provides a modern digital alternative.

Traditional Drawbacks

- ✗ Manual ledger entries prone to errors
- ✗ No GPS verification of pickup events
- ✗ Difficult to detect waste leakage in transit
- ✗ Compliance audits take days to reconcile
- ✗ Paper receipts get damaged or lost

The EcoTrace Standard

- ✓ Automated unique QR-barcode printing
- ✓ GPS verified coordinates at check-in
- ✓ Bluetooth scale integration for auto weight
- ✓ Instantly reconciled transit metrics
- ✓ Secure, immutable digital certificates

Guaranteed Chain of Custody

EcoTrace enforces a strict ****digital handshake**** at every handover node. From the moment the healthcare facility labels a waste bag, through transit check-in, and final treatment processing, every action is timestamped, scanned, and verified. Discrepancies are flagged immediately, preventing environmental leaks and ensuring absolute regulatory alignment.

Designed for Collaboration

EcoTrace is structured into three dedicated modules, delivering targeted, high-performance interfaces for all stakeholders in the waste lifecycle.

1

Healthcare Facility Portal (HCF)

Hospitals generate waste bags inside the portal, print dynamic color-coded QR labels (Yellow, Red, Blue, White), review dispatch manifests, and download official disposal certificates once processing completes.

2

Driver App (Mobile Optimized)

A responsive web application enabling field collectors to perform GPS check-ins at facility nodes, scan QR labels with their phone cameras, record weights directly via Bluetooth scales, and sync queue data offline.

3

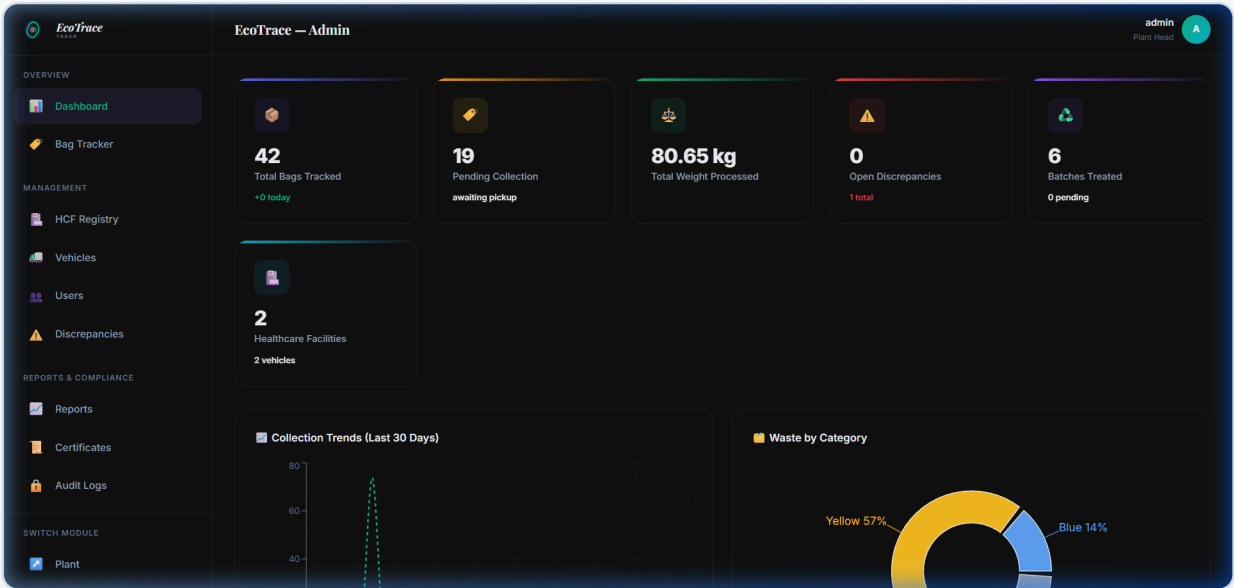
Common Treatment Facility Dashboard (CBWTF)

Plant managers log gate receipts, reconcile inbound shipments against driver route sheets, group bags into processing batches (Autoclave, Incinerate), and issue cryptographically backed disposal certificates.

Centralized Operations Hub

The EcoTrace Admin Dashboard provides facility heads with an overview of active routes, critical compliance discrepancies, and historical waste volumes.

Key Features: Recharts integration for visual weight distribution trends, live route sync monitoring, and HCF registry maps.



End-to-End Item Tracking

Every waste bag generated by an HCF receives a unique tracking profile. Handover status update events (Created, Transit, Received, Treated) are recorded instantly.

Key Features: Unique barcode parsing, color-coded categorization filters, and batch creation reconciliation indicators.

EcoTrace

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SWITCH MODULE

Plant

EcoTrace — Admin

admin

Plant Head

Bag Tracker

Create Bags

SEARCH

Barcode or hospital...

STATUS

All

CATEGORY

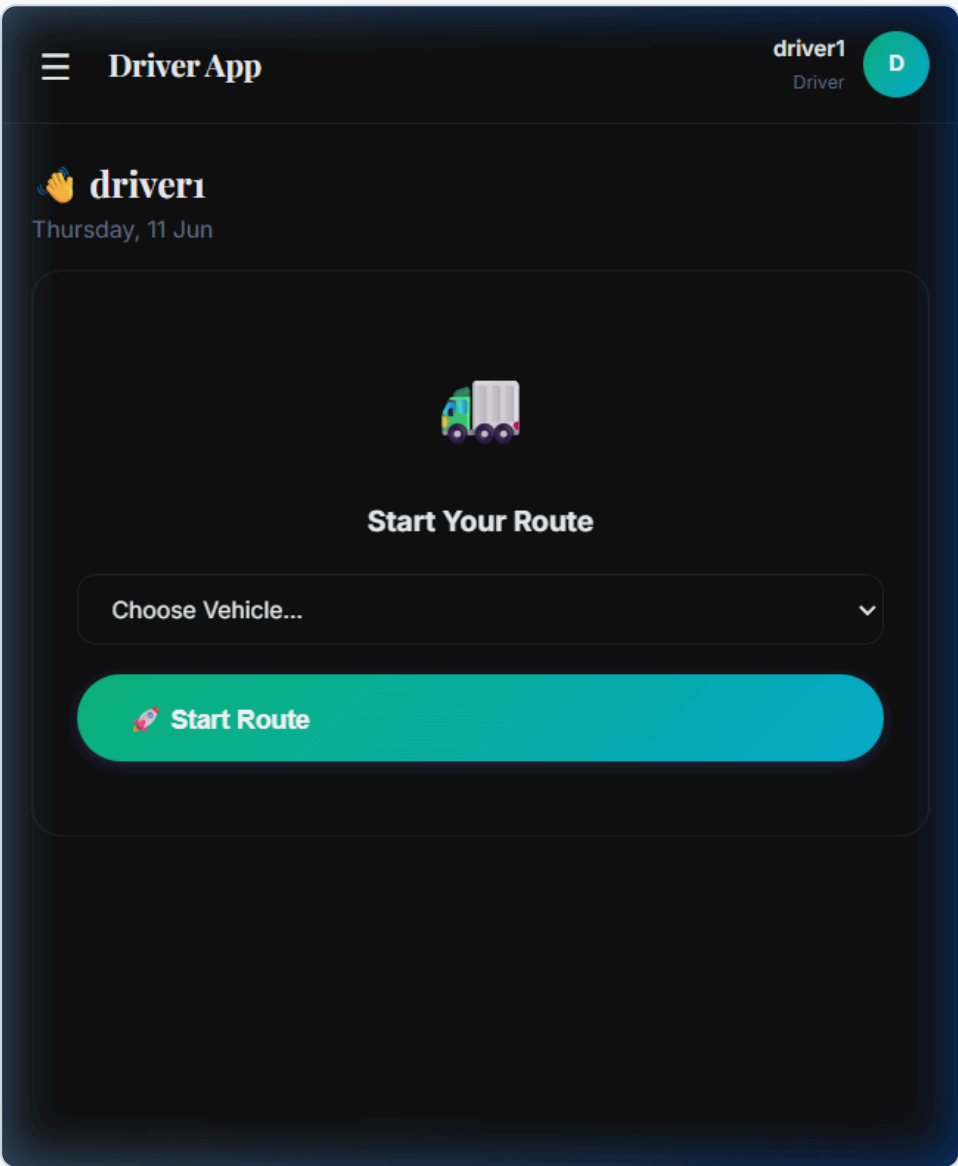
All

BAG ID / BARCODE	HOSPITAL	CATEGORY	WEIGHT	STATUS	CREATED	ACTIONS
3H-DUP-PMP/DUP/MAR1/26-27/00369-M-20260605-000005	Marik Polyclinic	White	—	created	5/6/2026	View
3H-DUP-PMP/DUP/MAR1/26-27/00369-M-20260605-000006	Marik Polyclinic	White	—	created	5/6/2026	View
3H-DUP-PMP/DUP/MAR1/26-27/00369-B-20260605-000004	Marik Polyclinic	Blue	3.1 kg	collected	5/6/2026	View
3H-DUP-PMP/DUP/MAR1/26-27/00369-B-20260605-000003	Marik Polyclinic	Blue	—	created	5/6/2026	View
3H-DUP-PMP/DUP/MAR1/26-27/00369-Y-20260605-000002	Marik Polyclinic	Yellow	1.75 kg	collected	5/6/2026	View
3H-DUP-PMP/DUP/MAR1/26-27/00369-Y-20260605-000001	Marik Polyclinic	Yellow	2.14 kg	collected	5/6/2026	View
3H-DEO-001-Y-20260605-000005	megdha seva sadan	Yellow	—	created	5/6/2026	View

On-the-Road Field Operations

Drivers handle collections via mobile check-ins and built-in camera scanning. Collected items are saved locally and synced automatically as connectivity changes.

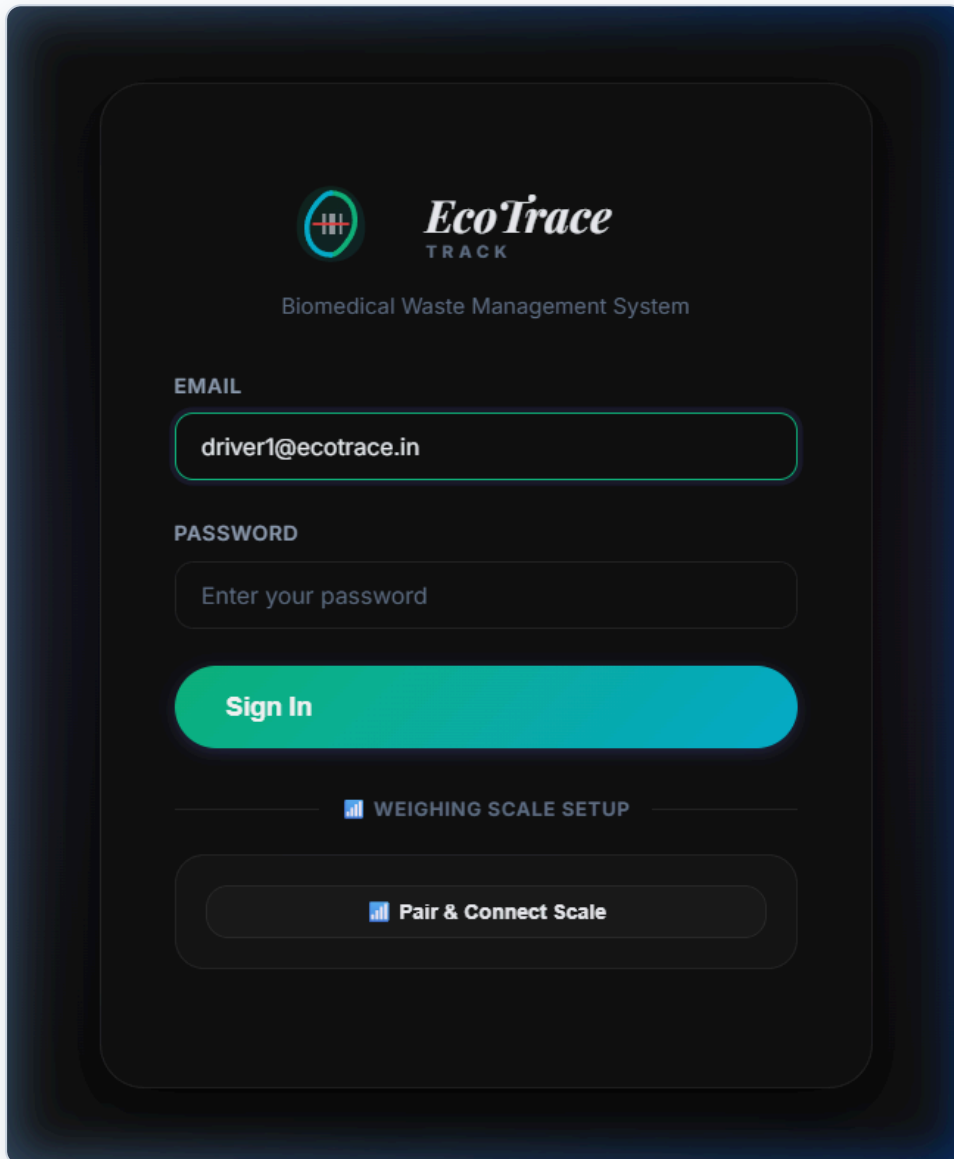
Key Features: Offline sync queues, real-time online/offline banners, HTML5 QR camera decoding, and Bluetooth weight scale pairing.



Direct Digital Weight Capture

To prevent clerical errors and compliance fraud, EcoTrace connects directly to digital weighing scales using Web Bluetooth APIs.

Key Features: Pair & Connect Scale widget, live weight streaming in kg, and automatic connection state persistence.



The image shows a dark-themed user interface for the EcoTrace Biomedical Waste Management System. At the top, the EcoTrace logo and name are displayed, followed by the text "Biomedical Waste Management System". Below this, there is a login section with fields for "EMAIL" (containing "driver1@ecotrace.in") and "PASSWORD" (with a placeholder "Enter your password"). A large green "Sign In" button is positioned below the password field. Further down, a section titled "WEIGHING SCALE SETUP" is visible, featuring a button labeled "Pair & Connect Scale".

Instant PDF Receipts & Auditing

Once waste is processed at the treatment plant, EcoTrace generates a digital Disposal Certificate detailing total weight and category breakdown.

Key Features: PDF generation engine, searchable certificate archive table, and direct print functionality for SPCB audits.



EcoTrace

Trace

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EcoTrace — Admin

admin

Plant Head

Treatment Certificates

Filter by HCF:

All Facilities (Batch-wise)

BATCH / HCF	BAGS	WEIGHT	TREATMENT	TREATED AT	ACTIONS
BATCH-1779879841898	3	4.90 kg	Autoclave	5/18/2026	View Certificate
BATCH-1779879827779	2	713 kg	Autoclave	5/18/2026	View Certificate
BATCH-1779879767043	4	53.80 kg	Autoclave	5/18/2026	View Certificate
BATCH-1779879672569	10	6.42 kg	Incineration	5/18/2026	View Certificate

Generate monthly waste output reports automatically. Visualize historical weight logs, hospital distributions, and segregation indexes.

Key Features: Interactive Recharts charts (area charts and bar charts), export to CSV/Excel capabilities, and monthly filter models.



EcoTrace

Waste Management

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Biomedical Waste Reports

PERIOD

MONTH

DISTRICT

HCF

HCF TYPE

Month-wise

June, 2026

All Districts

All HCFs

All

Biomedical Waste Summary Report

Month: 2026-06

HCF NAME	TYPE	BEDDED	YELLOW BAGS				RED BAGS				GEN BAGS	GEN WT(KG)
			GEN BAGS	GEN WT(KG)	REC BAGS	REC WT(KG)	GEN BAGS	GEN WT(KG)	REC BAGS	REC WT(KG)		
megdha sevo sadan	Private	Bedded	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000
Marik Polyclinic	Other	Non-Bed	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000
Grand Total			0	0.000	0	0.000	0	0.000	0	0.000	0	0.000

To maintain trust and security, EcoTrace features database-level row audit triggers that record every single profile action.

Key Features: Immutable PostgreSQL log ledger (no edit or delete policies), username tracking, action detail logging, and date-time timestamps.

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Plant Head

Audit Logs

Immutable — No deletion allowed

SEARCH

Search actions, users, details...

ACTION	USER	ENTITY	DETAILS	TIMESTAMP
HCF_DELETED	admin	HOSPITAL	Deleted HCF: abc hosp	6/9/2026, 12:06:21 AM
BAGS_CREATED	admin	BAG	Created 2 White bags for Markk Polyclinic	6/5/2026, 12:37:40 PM
DRIVER_CHECKIN	driver1	CHECKIN	Driver GPS Check-in: 24.492192, 86.722314	6/5/2026, 12:16:16 PM
BAG_COLLECTED	driver1	BAG	Bag JH-DUM-PWM/DUM/MARI/26-27/00369-B-202606...	6/5/2026, 12:14:35 PM
DRIVER_CHECKIN	driver1	CHECKIN	Driver GPS Check-in: 24.492191, 86.722349	6/5/2026, 12:13:03 PM
DRIVER_CHECKIN	driver1	CHECKIN	Driver GPS Check-in: 24.492191, 86.722349	6/5/2026, 12:13:03 PM
BAGS_CREATED	admin	BAG	Created 2 Blue bags for Markk Polyclinic	6/5/2026, 12:08:15 PM
BAG_COLLECTED	driver1	BAG	Bag JH-DUM-PWM/DUM/MARI/26-27/00369-Y-202606...	6/5/2026, 12:08:46 PM

Built for Scale & Trust

Leveraging modern architecture to ensure fast updates, military-grade data security, and seamless integration with existing IT infrastructure.

Premium Tech Stack

Developed using top-tier frameworks to offer smooth page loads and zero downtime.

React 18

Vite Dev

Supabase DB

PostgreSQL

RLS Policies

JWT Auth

Regulatory Compliance

- ✓ Aligned with BMW Rules 2016
- ✓ Strict 4-color categorization rules
- ✓ Fully auditable ledger reports
- ✓ Real-time GPS log trail mapping

Upgrade Your Compliance Today

Deploy EcoTrace within your facility network to eliminate waste discrepancy leaks, digitize logs, and automate regulatory reports.

[REQUEST A DEMO](#)